

Alexandre Champagne-Ruel

NASA Postdoctoral Program Fellow

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Origins of Life • Astrobiology • Complex Systems

RESEARCH PROFILE

Physicist and NASA Postdoctoral Fellow studying why driven chemical systems become complex rather than settle at equilibrium. I use stochastic artificial-chemistry simulations and assembly theory to test how environmental structure and energy flow shape molecular complexity before Darwinian evolution, and I am extending the same questions to biological regulatory networks. The applied aim is an abiotic baseline for agnostic life detection.

ACADEMIC POSITIONS

NASA Postdoctoral Program Fellow
Arizona State University, Mathis Group

2025 – Current
Tempe, AZ, USA

EDUCATION

Ph.D. Physics

2025

Université de Montréal

Montréal, QC, Canada

- Thesis: *Spatiality in Prebiotic Evolution: Toward a Physics of the Emergence of Complexity*
- Advisor: Paul Charbonneau

M.Sc. Physics

2020

Université de Montréal

Montréal, QC, Canada

- Thesis: *From Game Theory to Exobiology: The Emergence of Cooperation as a Critical Phenomenon*
- Advisor: Paul Charbonneau

B.Sc. Physics

2018

Université de Montréal

Montréal, QC, Canada

B.Sc. Philosophy

2012

Université de Montréal

Montréal, QC, Canada

GRANTS & AWARDS

Only individual grants >\$5,000 are listed.

NASA Postdoctoral Program Fellowship NASA

146,496 USD

2025

Project: Mapping Molecular Complexity for Agnostic Life Detection

J. Armand Bombardier Scholarship Fondation Bombardier

10,000 CAD

2024

Globalink Research Award Mitacs Canada

6,000 CAD

2024

Google Cloud Research Grant Google

5,000 USD

2023 – 2026

J. Armand Bombardier Scholarship Fondation Bombardier

10,000 CAD

2023

J. Armand Bombardier Scholarship <i>Fondation Bombardier</i>	10,000 CAD	2022
Doctoral Scholarship <i>Fonds de Recherche du Québec</i>	70,000 CAD	2022
Excellence Award <i>Université de Montréal</i>	10,000 CAD	2018
Various Travel & Research Grants <i>15 awards</i>	~30,000 CAD	2011 – 2025

PUBLICATIONS

Accepted Articles

- A. **Champagne-Ruel**, C. P. Kempes, and C. Mathis, “Spatial Patterning and Selection: How the Environment Shapes Molecular Complexity”, [10.48550/arXiv.2509.04547](https://doi.org/10.48550/arXiv.2509.04547) (2025), (Accepted for publication in PNAS Nexus).

Published Articles

- S. Asche, C. Bautista, C. Blanco, D. Boulesteix, A. **Champagne-Ruel**, C. Mathis, O. Markovitch, et al., “What it takes to solve the origin of life: An integrated review. Part 1–Experimental methods and data repositories”, [10.1016/j.xcrp.2026.103212](https://doi.org/10.1016/j.xcrp.2026.103212) (2026).
- S. Asche, C. Bautista, C. Blanco, D. Boulesteix, A. **Champagne-Ruel**, C. Mathis, O. Markovitch, et al., “What it takes to solve the origins of life: An integrated review. Part 2: Theoretical methods and emerging trends”, [10.1016/j.xcrp.2026.103211](https://doi.org/10.1016/j.xcrp.2026.103211) (2026).
- S. Rajamani, C. Mathis, S. Asche, D. Boulesteix, C. Blanco, O. Markovitch, A. **Champagne-Ruel**, et al., “Better origin(s) of life research through collaboration across disciplines”, [10.1016/j.xcrp.2026.103220](https://doi.org/10.1016/j.xcrp.2026.103220) (2026).
- A. **Champagne-Ruel**, S. Zakaib-Bernier, and P. Charbonneau, “Diffusion and pattern formation in spatial games”, *Physical Review E* **110**, 014301 (2024).
- A. **Champagne-Ruel** and P. Charbonneau, “A Mutation Threshold for Cooperative Takeover”, *Life* **12**, 254 (2022).
- S. Gelin, A. **Champagne-Ruel**, and N. Mousseau, “Enthalpy-entropy compensation of atomic diffusion originates from softening of low frequency phonons”, *Nature Communications* **11**, 3977 (2020).

PRESENTATIONS

Invited Talks

Spatiality in Prebiotic Evolution <i>Queens College, City University of New York</i>	2026 <i>New York, USA</i>
Physics of Complexity And Agnostic Life Detection <i>Trottier Institute for Research on Exoplanets</i>	2025 <i>Montréal, QC, Canada</i>
Spatial Patterning and Selection <i>Max Planck Institute for Terrestrial Microbiology</i>	2025 <i>Marburg, Germany</i>
From Emergent Complexity to Reliable Life Detection <i>Arizona State University</i>	2024 <i>Tempe, AZ, USA</i>
Coopération, émergence et transitions <i>Université du Québec à Montréal</i>	2022 <i>Montréal, QC, Canada</i>

Contributed Talks

How Environments Shape Molecular Complexity: From Abiogenesis to Life Detection	2026
<i>Interdisciplinary Origin of Life Meeting (IOoL)</i>	<i>Tokyo, Japan</i>
Network Topology as a Tunable Parameter for Emergent Complexity	2026
<i>Conference on Complex Systems (CCS)</i>	<i>Binghamton, NY, USA</i>
From Emergent Complexity to Reliable Life Detection	2025
<i>Biennial European Astrobiology Conference (BEACON)</i>	<i>Reykjavik, Iceland</i>
Diffusion: An Overlooked Driver of Prebiotic Complexity	2024
<i>Astrobiology Science Conference (AbSciCon)</i>	<i>Providence, RI, USA</i>
Théorie de l'information et origine de la vie	2023
<i>90e Congrès de l'ACFAS</i>	<i>Montréal, QC, Canada</i>
A Mutation Threshold for Cooperative Takeover	2022
<i>Astrobiology Science Conference (AbSciCon)</i>	<i>Atlanta, GA, USA</i>
Cooperation: An Emergent Universal Feature at the Dawn of Life	2022
<i>Interdisciplinary Origin of Life Meeting for Early Career Researchers</i>	<i>Montréal, QC, Canada</i>
Mutation Favors the Emergence of Cooperation	2021
<i>Life and Space Poland</i>	<i>Virtual/Szczecin, Poland</i>
La criticité dans un système évolutif artificiel	2019
<i>Centre de recherche en astrophysique du Québec (CRAQ)</i>	<i>Orford, QC, Canada</i>

Posters

How Environments Shape Molecular Complexity: From Abiogenesis to Life Detection	2026
<i>Astrobiology Science Conference (AbSciCon)</i>	<i>Madison, WI, USA</i>
Mapping Molecular Complexity for Agnostic Life Detection	2025
<i>NASA Postdoctoral Program Virtual Symposium</i>	<i>Virtual</i>
L'émergence de la coopération via l'évolution de réseaux informationnels	2023
<i>90e Congrès de l'ACFAS</i>	<i>Montréal, QC, Canada</i>
Diffusion, structures spatiales et origine de la vie	2023
<i>90e Congrès de l'ACFAS</i>	<i>Montréal, QC, Canada</i>
OoLEN - The Origin of Life Early-career Network: Building the Community Needed to Solve the Problem	2022
<i>AbSciCon</i>	<i>Atlanta, GA, USA</i>
A Mutation Threshold for Cooperative Takeover	2022
<i>Gordon Research Seminar: Challenging Paradigms in Prebiotic Chemistry</i>	<i>Oxnard, CA, USA</i>
A Mutation Threshold for Cooperative Takeover	2022
<i>Gordon Research Conference: Environments for the Origins of Life and Habitability</i>	<i>Oxnard, CA, USA</i>
Les mutations favorisent la coopération en contexte évolutif	2021
<i>Centenaire, Département de Physique, Université de Montréal</i>	<i>Montréal, QC, Canada</i>
Mutation Favors the Emergence of Cooperative Behavior	2021
<i>Molecular Origins of Life Munich</i>	<i>Virtual/Munich, Germany</i>

Mutations Promote Cooperation in an Evolutionary Setting <i>XIXth ISSOL Conference</i>	2021 <i>Virtual</i>
Self-Organized Criticality : A Prelude to Avalanche Models of Solar Flares <i>Space Climate 7 Symposium</i>	2019 <i>Orford, QC, Canada</i>

SCHOOLS & SELECTIVE WORKSHOPS

Assembly Theory for Folded Matter <i>Santa Fe Institute</i>	2025 <i>Santa Fe, NM, USA</i>
Information Driven States of Matter <i>University of Rochester</i>	2024 <i>Rochester, NY, USA</i>

TEACHING EXPERIENCE

Mentor – Biocomputing Scholars program <i>Arizona State University</i>	2025 – 2026 <i>Tempe, AZ, USA</i>
Guiding 3 undergraduate students in developing computational chemistry simulations.	
Undergraduate Internship Supervision <i>Université de Montréal</i>	2022 <i>Montréal, QC, Canada</i>
Supervised a summer research project on diffusion; resulted in journal publication and poster presentation.	
Teaching Assistant – Introduction to Astrobiology <i>Université de Montréal</i>	2021 – 2022 <i>Montréal, QC, Canada</i>
Tutoring – Undergraduate Level <i>Université de Montréal</i>	2018 – 2022 <i>Montréal, QC, Canada</i>

SERVICE & OUTREACH

Scientific Leadership

Positions

Member of the Executive Board <i>Origin of Life Early-career Network (OoLEN)</i>	2022 – Present
Previously head of the virtual-meetings workgroup; organizer of the OoLEN Public Seminars lecture series (2023–24).	
Founder and Convener <i>cs-astrobio, international journal club and discussion group on complex-systems methods in astrobiology</i>	2026 – Present
Origin of Life Digest (link) <i>Editor</i>	2021 – Present

Conference Organization

Interdisciplinary Origin of Life Meeting <i>Organizing Committee</i>	2026 <i>Tokyo, Japan</i>
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Frontiers in Astrobiology and Origins of Life Conference

Organizing Committee

2025

Reykjavik, Iceland

Origine de la vie : de l'astrophysique à la philosophie

Lead Organizer

2023

Montréal, QC, Canada

Interdisciplinary Origin of Life Meeting for Early Career Researchers

Lead Organizer

2022

Montréal, QC, Canada

Space Climate 7

Local Organizing Committee

2019

Orford, QC, Canada

Session Convener

AbSciCon

Assembly Theory Across Scales: From Molecules to Planetary Systems

2026

Madison, WI, USA

AbSciCon

Exploring Self-Assembly and Self-Organization in Prebiotic Complexity

2026

Madison, WI, USA

Media & Public Engagement

É. Beaudoin-Paul, "Entrevue avec Alexandre Champagne-Ruel, récipiendaire de la prestigieuse bourse en astrobiologie de la NASA", [Quartier Libre](#) (2025).

A. Riopel, "Comment reconnaître la vie sur d'autres planètes", [Le Devoir](#) (2023).

C. Tamura, "Cooperation and the Origin of Life", Quantum Photonics Clubhouse Podcast (2022).

Reviewing Activities

- External Reviewer, NASA Solar System Science Program (2026)
- Journal of the Royal Society Open Science
- Journal of the Royal Society Interface
- Trends in Biochemical Sciences

Memberships

- International Society for Artificial Life
- Center for Research in Astrophysics of Québec
- Canadian Association of Physicists
- Canadian Astronomical Society
- Origin of Life Early-career Network
- International Society for the Study of the Origin of Life
- Complex Systems Society
- Scientific Society for Astrobiology (Founding Member)

TECHNICAL SKILLS

- **Languages:** Fluent in Spoken/Written French, English
- **Programming:** Python, C++, Fortran, Julia, R, LaTeX, MATLAB, x86 assembly, Bash
- **Modeling:** Agent-based, Evolutionary Algorithms, Machine Learning, Network Theory, Game Theory
- **Development Workflow:** Linux, High Performance Computing (HPC)/SLURM, Git, Jekyll, Hugo, HTML5/CSS3